Experimental Analysis of the Combustion Behavior of a Gas Turbine Burner by Laser Measurement Techniques

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Experimental results from optical and laser spectroscopic measurements on a scaled industrial gas turbine burner at elevated pressure are presented. Planar laser induced fluorescence on the OH radical and OH chemiluminescence imaging were applied to natural gas/air flames for a qualitative analysis of the position and shape of the flame brush, the flame front and the stabilization mechanism. The results exhibit two different ways of flame stabilization, a conical more stable flame and a pulsating opened flame. For quantitative results, one-dimensional laser Raman scattering was applied to these flames and evaluated on an average and single-shot basis in order to simultaneously determine the major species concentrations, the mixture fraction, and the temperature. The mixing of fuel and air, as well as the reaction progress, could thus be spatially and temporally resolved, showing differently strong variations depending on the flame stabilization mode and the location in the flame.* [DOI: 10.1115/1.3205033]

Keywords: laser diagnostics, Raman scattering, gas turbine combustor, flame stabilization

1 Introduction

The lean premixed combustion concept is widely applied in stationary gas turbines (GTs) in order to achieve low NO_x levels. However, these flames are prone to combustion instabilities in the form of unsteady flame stabilization or thermoacoustic pulsations [1–5]. These phenomena arise from a complex interaction between the turbulent flow field, the flame reactions, and the combustor geometry. Furthermore, effects of unmixedness have a significant influence on the flame stabilization and emissions [6,7]. Different approaches have been reported to reduce or even to eliminate the unsteady effects, e.g., active or passive control mechanisms [8–13] or fuel staging [14]. However, a fundamental understanding of the underlying mechanisms has not been reached so far, and further experimental and numerical studies are necessary before the potential of lean premixed combustion can be fully exploited.

In recent years, detailed laser diagnostic studies in GT-relevant combustors have contributed significantly to a better understanding of processes such as flame stabilization, combustion instabilities, pollutant formation and finite-rate chemistry effects. Frequently used measuring techniques were particle image velocimetry (PIV) or laser Doppler velocimetry (LDV) for the flow field; laser induced fluorescence (LIF) for flame radicals, tracers, pollutants, or temperature; coherent anti-Stokes Raman scattering (CARS) for temperature; and laser Raman scattering for species concentrations. Overviews and detailed descriptions of the different laser measurement techniques can be found in Refs. [15–18]. In some research groups, high-pressure test rigs have been equipped with optical access in order to apply these techniques to GT combustion processes under realistic conditions, i.e., preheated air, elevated pressure, high thermal loads, high turbulence levels, and relevant combustor geometries, see for example Refs. [19–28]. Here, challenges arise from the limited optical access and especially from window contamination by the high thermal loads or particles in the flow [29]. Particularly for single-shot one-dimensional (1D) Raman measurements, which require high laser pulse energies, a high laser fluence can lead to window damage or optical breakdown in the focal region.

In the investigations presented here, a scaled industrial GT-burner was installed in the high-pressure test rig at the Institute of Combustion Technology of the German Aerospace Center (DLR) in Stuttgart and investigated using different laser measuring techniques in combination with chemiluminescence imaging, exhaust gas analysis, and dynamic pressure recording. The combustor was operated with natural gas (NG) and preheated air ($T=400\,^\circ\mathrm{C}$), and the flame settings were varied in a wide range of thermal power, equivalence ratio, and fuel staging. Main goals of the study were a detailed characterization of the combustion process, the influence of fuel staging on the flame stabilization, the quantification of the degree of mixing, the identification of effects of turbulence-chemistry interactions, and the investigation of transient flame behavior. Planar laser induced fluorescence (PLIF) of OH and OH* chemiluminescence imaging were applied to determine the flame structures and the stabilization region. Further, 1D-laser Raman scattering was used to quantitatively measure the major species concentrations (N_2, O_2, H_2O, C_xH_y, CO_2, CO, and H_2) and the temperature. By this technique, the local mixture fraction and the reaction progress could be determined, revealing effects of unmixedness and turbulence-chemistry interactions.

The focus of this paper lies on the comparison of two different flames in terms of their shape and stabilization on the burner. Detailed quantitative results from single-shot 1D-Raman measurements on a scaled industrial GT-burner under realistic conditions are presented for the first time to our knowledge.

2 Experimental Setup

2.1 Burner and Test Rig. The experiments were conducted on a geometrically downscaled version of a gas turbine burner

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Contributed by the International Gas Turbine Institute of ASME for publication in the Journal of Engineering for Gas Turbines and Power. Manuscript received April 8, 2009; final manuscript received May 14, 2009; published online March 5, 2010. Review conducted by Dilip R. Ballal. Paper presented at the ASME Gas Turbine Technical Congress and Exposition, Orlando, FL, June 8–12, 2009.

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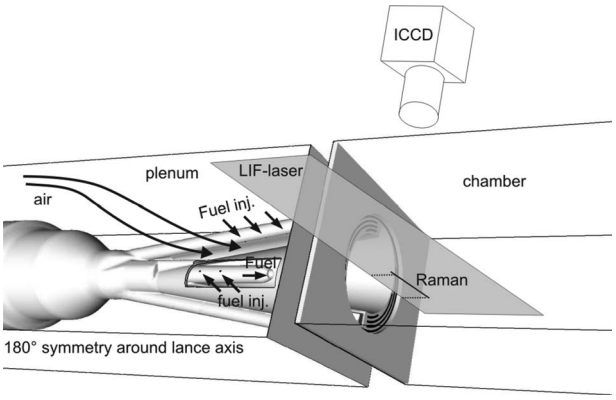

Fig. 1 Schematic drawing of the burner and the measurement positions of the different laser diagnostic techniques

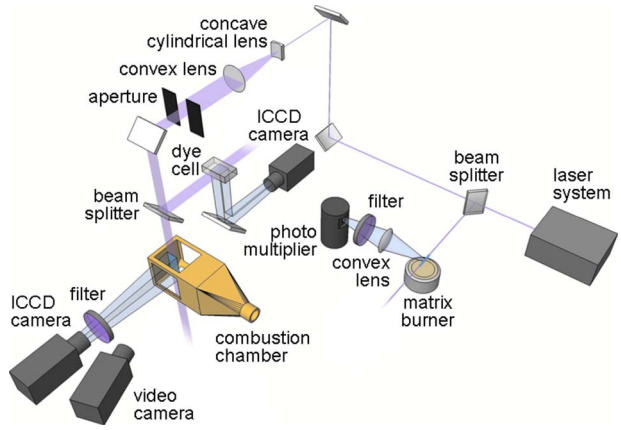

Fig. 2 Schematic drawing of the experimental setup for OH-PLIF measurements

(Alstom EV burner) [30,31]. The burner has already been the object of various experimental and numerical investigations with respect to the flow field, the flame behavior, emissions, the flame stability, and the mechanism of flame stabilization, see Refs. [14,32]. A schematic drawing of the burner is shown in Fig. 1. The positions of the laser sheet for LIF measurements and the radial profiles where the Raman measurements were taken are also indicated. It should be noted that the laser beam for the 1D-Raman measurements was directed orthogonally to the LIF laser sheet; the line in Fig. 1 refers to the middle of the imaged range, which intersects the LIF sheet.

The design of this swirl-stabilized burner is based on two halves of a cone, which are radially shifted apart. This generates two tangential slots of constant width, through which the combustion air enters the cone. By this concept, a tangential velocity component is induced in the flow, with an increasing swirl strength downstream and which results in a vortex breakdown close to the burner exit ($D=70$ mm). The fuel is injected through small holes along the tangential slits and mixes with the entering air. Additional fuel can be injected through the tip of a pilot lance, which protrudes on the axis 54 mm upstream from the burner mouth. The two fuel flows (tangential slots and pilot lance) can be controlled separately for different fuel staged operation conditions. The staging ratio is defined as the ratio of the fuel mass flow through the lance to the total fuel mass flow. For the present study, the EV type burner was operated and investigated in the high-pressure combustion test facility HBK-S (Hochdruckbrennkammerprufstand-Stuttgart) at the DLR Stuttgart. The test rig and the specially designed rectangular combustion chamber (99×129 mm^2) were both equipped with large quartz glass windows to provide full optical access on all four sides. The outer windows in the pressure vessel were exposed to elevated pressure but only at intermediate to low temperatures, while the inner ones were directly exposed to the flame but only moderate pressure variations. The inner windows were installed in a double-glazed configuration with cooling air flowing between the glass plates in order to reduce the thermal load of the windows. So, there were three windows in total both on the laser entrance and the exit side, as well as on the two sides in line of the detection. The burner plate was cooled by air streaming through small holes into the combustion chamber.

2.2 Flames. The burner was operated with natural gas/air flames with a thermal power up to 370 kW at different values of air equivalence ratio λ and staging ratios. The NG was taken from the public power supply. The composition of the NG hardly varied during the period of the measurement campaign with a mean composition of 94.14% CH_4, 2.955% C_2H_6, 0.557% C_3H_8, 0.064% C_4H_{10}, 0.07% isobutane, 2.124% N_2, and 0.09% CO_2. The air preheat temperature was kept constant to 673 K, the pressure inside the combustion chamber was 3 bars. It has been observed in the flow field that the burner can exhibit two different ways of flame stabilization, depending on the several conditions, as it has been reported from several investigations, both experimental and theoretical [14,32,33]. One is a conical and more stable flame with respect to pulsations, which is anchored inside the burner cone. The other one is stabilized just downstream the burner exit. In this case, the flame is more opened and the outer recirculation zone (ORZ) is more pronounced compared with the cone stabilized flame. The opened mode has better mixing leading to lower NO_x, but it is also more prone to pulsations. Generally speaking, the transition from the cone stabilized to the opened flame takes place with increasing λ (decreasing reactivity) with a relatively small range in which the flame alternates between the two states, or with a reduced staging ratio. The transition to the opened flame is accompanied by a reduction of NO_x emissions and a pressure drop across the burner, both in the case of increasing λ and the case of a reduced staging ratio. In this study, the results of 1D-Raman measurements of two selected flames are presented, one of each way of stabilization, and compared with the according LIF and OH^* chemiluminescence measurements. The two flames are referred to as flames A and B in this paper. Flame A is a cone stabilized flame with a thermal power of 345 kW, a staging ratio α_A of 0.23, and $\lambda_A\approx1.3$. Flame B is an opened flame with 317 kW, $\alpha_B=0.17$, and $\lambda_B\approx1.7$. The corresponding mixture fractions f are $f_A=0.044$ and $f_B=0.034$ for flames A and B, respectively.

2.3 PLIF/OH* Chemiluminescence Imaging. For the OH-PLIF measurements a flashlamp pumped frequency doubled Nd:YAG laser (Quanta Ray, DCR-2) was used to pump a frequency doubled tunable dye laser (Lumonics, HD-500) at 10 Hz. The UV beam (pulse duration of 8 ns and line width of 0.4 cm^{-1}) had a pulse energy of approximately 2.5 mJ at the exit of the laser and was tuned to the $Q_1(8)$ transition of OH at approximately 283 nm in the $\nu''=0\rightarrow\nu'=1$ vibrational band of the A $^2\Sigma^+$-X$^2\Pi$ system. The population of the laser-coupled ground state of the selected line varied by approximately 9% in the temperature range of interest (1400–2200 K). Subsequent OH fluorescence in the $\nu'=1\rightarrow\nu''=1$ and $\nu'=0\rightarrow\nu''=0$ bands near 310 nm was detected through an interference filter in the wavelength region 295–340 nm. This enabled suppression of light scattered from the incident laser beam and background radiations. The output beam from the dye laser was expanded into a vertical sheet by means of cylindrical lenses, as shown in Fig. 2. The laser sheet inside the combustor was 60 mm in height and approximately 400 μm in thickness. By means of a beam splitter, a small portion of the incoming laser beam was directed into a CH_4/air reference flame on a matrix burner, operated under laminar conditions. The LIF produced

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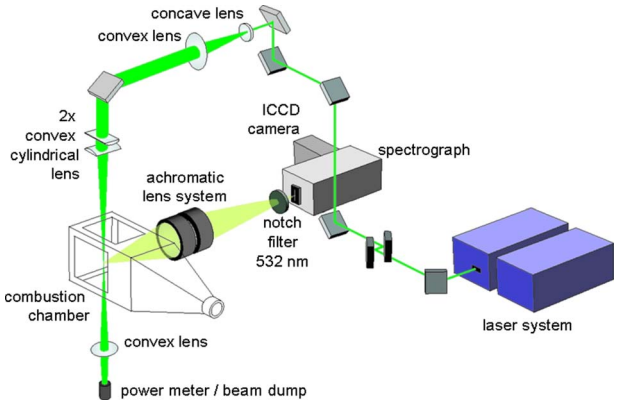

Fig. 3 Schematic drawing of the setup for 1D-Raman experiments

by that flame was used for the online monitoring of the excitation line wavelength.

The OH-PLIF detection system consisted of an image-intensified charge coupled device (ICCD) camera (Roper Scientific, 512×512 pixels) with achromatic UV lens (Halle, $f/2$, $f=100$ mm). Part of incoming laser sheet was deflected into a quartz cell filled with fluorescent dye solution by means of a beam splitter. The dye cell fluorescence profile was simultaneously imaged using another ICCD camera (Roper Scientific, 512×512 pixels) equipped with a Nikon lens ($f/4.5$, $f=105$ mm). This dye cell profile was used for correcting the laser sheet profile inhomogeneities. The image intensifier was set to an exposure time of 400 ns for both the OH-PLIF and the dye cell fluorescence camera. In this experiment, an arrangement for the correction of absorption [34,35] was not applied. The same detection system along with the filter combination was used for the OH^* chemiluminescence measurements. The exposure time for the image intensifier was set to 40 μs to capture the integrated (along the line of sight) spontaneous emission from OH in the electronically excited state (OH^*). The timing between the laser pulses and the camera gate openings were synchronized by means of a pulse delay generator (DG535, SRS Inc., Sunnyvale, CA).

2.4 1D-Laser Raman Scattering. Single-shot vibrational Raman scattering was applied to simultaneously determine the major species concentrations (N_2, O_2, H_2O, NG, CO_2, CO, and H_2) along a line 7 mm in length. The NG was treated as one species in the detection and evaluation procedure. The various hydrocarbons within the NG all contribute to the Raman-active C–H stretch vibration, which was detected at around 630 nm. The inaccuracy associated with this simplification is relatively small because the maximum NG mole fractions measured in the investigated premixed flames were approximately 0.1. Thus, the mole fractions of the minor constituents of the NG are below 0.01. The main components of the experiment were a laser system consisting of five Nd:YAG lasers, a pulse stretcher, beam forming optics, the detection optics, a spectrograph, and an ICCD camera, as shown in Fig. 3. The Nd:YAG lasers provided five temporally separated pulses at a wavelength of $\lambda=532$ nm of 7 ns duration, each at a repetition rate of 10 Hz, which were combined to one beam [36]. In order to reduce the intensity at the windows and to avoid optical breakdown in the focal region the pulses were elongated by a pulse stretcher, and furthermore the laser beams were focused by a combination of two perpendicularly arranged cylindrical lenses ($f=300$ mm) to generate a "blurred" focus of an average diameter of 0.5 mm [29]. In this way, a total pulse energy of 0.8 J could be irradiated into the combustor. The pulse stretcher consisted of two beam splitters and six mirrors, which divided each pulse into several individual pulses, which were recombined again after having traveled different path lengths, generating a total pulse duration of about 350 ns [37]. The Raman scattered light was collected at 90 deg by a custom designed achromatic lens system, with an aperture of 150 mm configured for line imaging of Raman scattering at 532 nm [36] and relayed onto the entrance slit of the spectrograph (SpectraPro 300i, Acton Research, $f=300$ mm, 490 lines/mm, $f_{\#}=4$, and dispersion ~6 nm/mm). For the rejection of elastically scattered light, a holographic notch filter was placed in front of the entrance slit. A slit width of 1 mm, a slit height of 14 mm, and a magnification of the detection optics of 2 resulted in an imaged probe volume of 7 mm in direction of the laser beam and 0.5 mm in diameter. After spectral separation, the Raman bands from the different major species were detected by an intensified CCD camera (1340×1300 pixels) along the line of 7 mm. The images were binned on chip to superpixels (268 in the direction of the spectral separation and 28 in spatial direction). This yielded a partition of 28 measurement volumes along the laser line with 0.25 mm length. The laser was introduced from the top of the combustor, and the focus was always kept in the middle of the combustion chamber in vertical direction. The measuring location within the combustion chamber was changed in the direction of the detection line by translating the laser beam optics and detection optics simultaneously via computer controlled translation stages. Due to restrictions of the optical access, the measuring location closest to the burner mouth was 22 mm. For the measurements presented in this paper, 500 single shots were recorded per location.

One of the main challenges in performing the Raman measurements was the degradation of the combustor windows, which were directly exposed to the flame. At high flame temperatures, quartz begins to soften and its surface becomes rough and increasingly opaque leading to decreasing transmittance and, in the worst case, to window damage by the laser beam. The degradation was not uniformly distributed over the window and most severe where the flame hit the quartz surface. As a result, the effective laser pulse energy at the measuring location could not always be determined accurately. However, this quantity is usually needed to ratio the Raman signals in order to determine the species concentrations quantitatively. A similar uncertainty arose from the unknown (time and position dependent) transmission of the window on the detection side. Therefore, it was necessary to find an appropriate method to compensate for the lacking information.

2.5 Raman Data Evaluation. In order to deduce the species number densities from the Raman signals, calibration measurements were performed. Temperature dependent calibration and crosstalk factors were determined in laminar flat flames (1150–2200 K) with known temperature and gas compositions operated at atmospheric conditions in the GT combustion chamber. The coefficients for the low temperature range ($T=300-1000$ K) were determined in cold and electrically heated gas flows. The temperature is deduced from the total number density using the ideal gas law and the measured pressure in the combustion chamber. The measured signals from the GT flames were evaluated using these calibration curves. Because the signals have to be weighted by the laser pulse energy (on a single-shot basis), the uncertainty of the determination of the laser energy caused by degraded windows led to a significant uncertainty in the determination of the species number densities. Although the relative species composition was hardly affected by this uncertainty, the total number density and thus the temperature are. Therefore, the following procedure was applied to determine the temperature. All C and H atoms in the combustion chamber stem from the fuel and all N and O atoms from the air supply (except for a negligible contribution from the natural gas). During all processes in the combustion chamber (transport and chemical reactions) the atomic ratios C/H and N/O remain unaltered in every volume element, i.e., 0.26 for C/H and 3.73 for N/O. A correct Raman measurement must therefore yield these numbers. On the other hand, the Raman calibration curves for the different molecular

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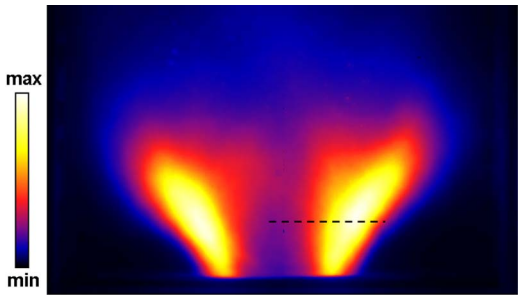

Fig. 4 Mean distribution of the OH* chemiluminescence of flame A (averaged over 200 single shots). The dotted line indicates the position of the radial profile that was measured by 1D-Raman scattering.

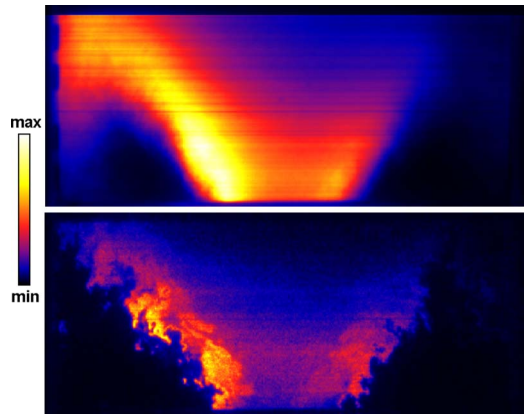

Fig. 5 OH-PLIF measurements of flame A. Top: mean OH distribution (averaged over 200 single shots). Bottom: single-shot OH-PLIF measurement.

species exhibit different temperature dependencies. Fortuitously, the temperature dependencies vary in a way that the C/H and N/O ratios deduced from the measurements are a unique and noticeable function of T. Taking the relative Raman signal intensities from each single-shot measurement, it turned out that only for one temperature value the ratios C/H and N/O assumed the expected values. The Raman-spectrum was therefore scaled to match the optimum value of both the C/H and the N/O ratio. The species number density and the temperature were then calculated based on the scaled spectrum.

The precision of the Raman results can be estimated by the single-shot evaluation and shot noise analysis of measurements on the stable calibration flames. The relative standard deviations in the postflame region of a flat CH_4/air flame at atmospheric pressure are typically around 4% for N_2, 13% for H_2O, 20% for CO_2, and 7% for the temperature and the mixture fraction. The absolute temperatures of the calibration flames were reproduced by an accuracy better than 5%. The flame temperatures had been determined beforehand using CARS with an accuracy of 3%. The temperature data for flat calibration flames obtained by CARS measurements have been published in Refs. [38,39]. As the Raman signals scale linearly with pressure, the signals from the GT flames (3 bars) are a factor of 3 higher than in the calibration flames. The precision should therefore be improved by a factor of $\sim\sqrt{3}$. However, systematic uncertainties due to shifts in alignment or signal correction might impair the accuracy.

3 Results

3.1 Cone Stabilized Flame A. The measurement results of the two different flames are presented one after another. First, the cone stabilized flame A is discussed. In Fig. 4 the mean OH* chemiluminescence of this flame is shown, averaged over 200 single shots. The electronically excited OH is formed by the flame reactions in the region of the highest heat release rates and is therefore a good marker for the position of the flame front. In all shown pictures, the flow is from the bottom up. One can clearly see that the flame brush has a conical shape and is mainly concentrated in a distinct layer in the outer region of the swirled flow exiting the burner. The combustion of the fresh gas already takes place inside the burner and just outside the burner exit. One should keep in mind, however, that the detected signal of the chemiluminescence is integrated over the whole flame in the line of sight and gives no information about the flame structure details with an appropriate spatial resolution. The slight asymmetry in the intensity distribution might partly be due to inhomogeneities in the transparency of the combustor windows. However, the combustor geometry is not axial symmetric, so that the observed asymmetry of the flame might be real.

The PLIF measurements on OH reveal the hot regions in the combustion chamber where the OH radical is present ($\geq$1400 K). The results of the PLIF measurements of flame A are shown in Fig. 5. On top, the mean OH distribution is shown, averaged over 200 single shots. The asymmetry is caused by absorption of the laser radiation by OH radicals and thus attenuation of the laser light sheet, which enters the combustion chamber from the left side. The hot inner recirculation zone (IRZ) reaches back into the burner, where it promotes the ignition of the fresh gas and stabilizes the flame. The bottom picture shows a single shot of the OH-PLIF measurements. Like the OH* chemiluminescence measurements, it also shows the conical shape of the flame and reveals a strongly wrinkled flame front due to the turbulence of the flow. Regions of high signal intensity (red and yellow) indicate the location of the flame front, where OH is formed in superequilibrium concentrations [40].

Such a qualitative analysis of this flame by OH-PLIF and OH* chemiluminescence has already been reported elsewhere [14]. The new component in the present study is the successful application of laser Raman scattering, which allows for a quantitative analysis with respect to the major species, the temperature, and the mixture fraction. The results of the evaluated mixture fractions f in flame A at different radial positions 22 mm downstream are shown in Fig. 6. The symbols represent the ensemble averaged values of the measuring volume in the plane of the PLIF laser sheet over 500 single shots. As expected for a technically premixed flame, the mixture fraction is not constant, however, the variation in the mean values is rather moderate over a large region. Close to the burner axis (-5 mm$\leq r \leq 5$ mm), the mixture fraction is slightly increased and asymmetric, which is presumably due to the asymmetry of the flow field, as the burner is not rotationally symmetric.

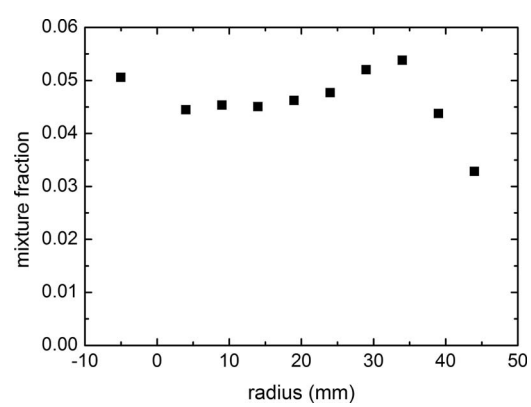

Fig. 6 Radial profile of the mean mixture fraction in flame A 22 mm downstream from the burner mouth; for comparison, the stoichiometric mixture fraction of the NG is 0.057

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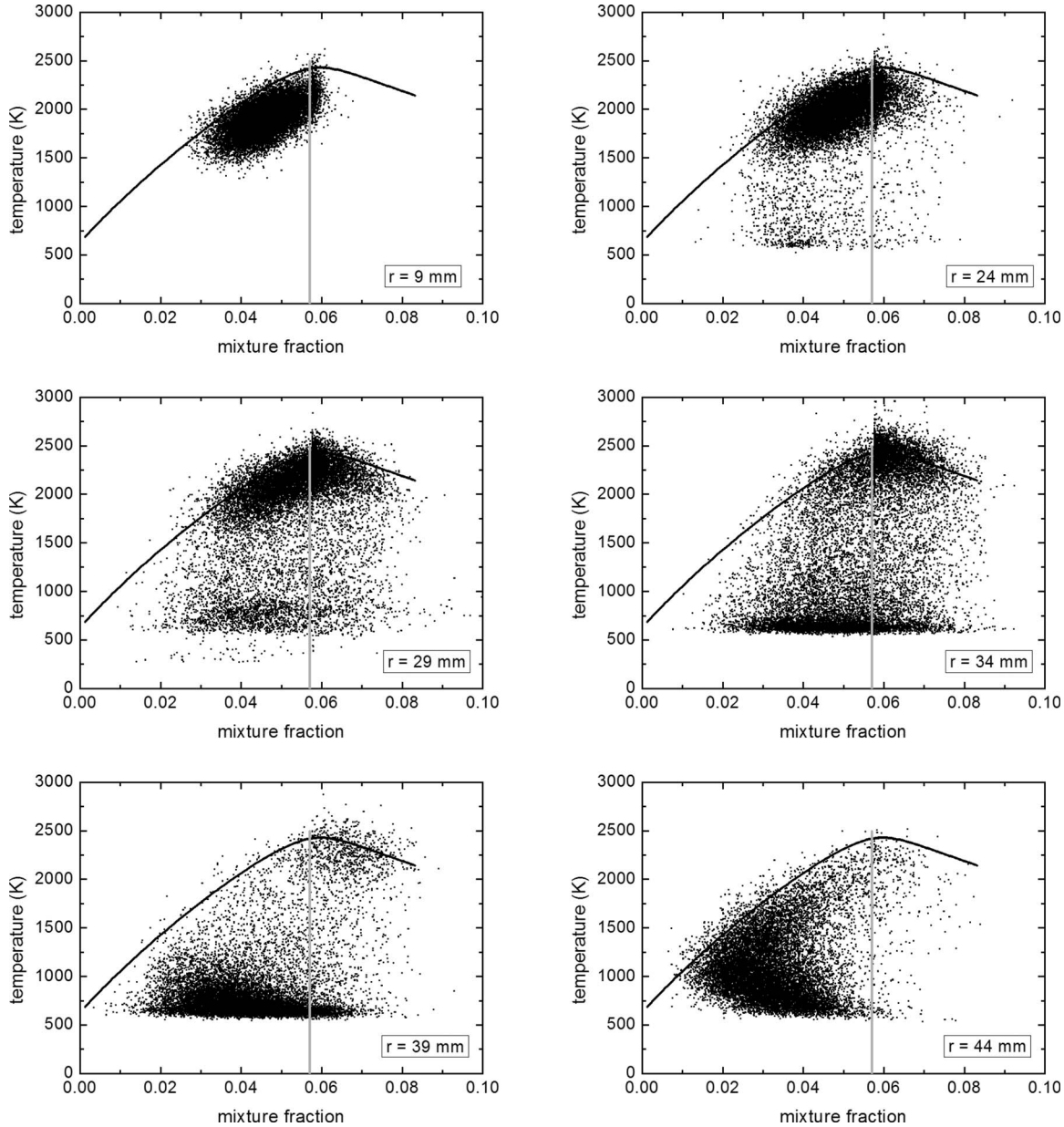

Fig. 7 Scatter plots of single-shot Raman measurements in flame A at different radial positions 22 mm downstream from the burner mouth. Adiabatic equilibrium (black) and stoichiometric mixture fraction (light gray) are indicated by lines.

From $r=5$ mm outwards, the mixture fraction remains almost constant until it starts to increase at a radius of $r=25$ mm. It reaches the maximum value at $r=34$ mm, which is also the region where the highest signal intensities can be found in the PLIF and OH^* chemiluminescence measurements at this distance from the burner mouth. Farther outwards the mixture fraction strongly decreases, which can be explained by the hot gas in the outer recirculation zone mixing with the cooling air for the burner plate, which streams into the combustion chamber through the burner plate. In Fig. 7 scatter plots are displayed showing the correlation between the temperature and mixture fraction at some of the prior shown radial positions at the same downstream location (22 mm). The data points represent the results of the single-shot measurement from all 28 measuring volumes along the laser line. In addition, the indicated black curve shows the correlation at adiabatic equilibrium conditions, which was calculated with GASEQ [41]. It is not expected that the investigated flame is in a state of adiabatic equilibrium, however, the curve helps to assess the thermochemical state of the flame. The light gray vertical line indicates the stoichiometric mixture fraction for the given NG composition. In general, the scatter plots reflect a significant variation in f, which changes with the radial position. For an industrial GT-burner, it is the first time that this variation could be determined quantitatively on a single-shot basis, i.e., without temporal or spatial averaging. The second general feature, which becomes immediately obvious from the scatter plots, is the significant variation in temperatures reflecting different states of reaction progress. In the following the different radial positions are discussed in more detail. At $r=9$ mm, the temperatures are high with little fluctuations and the samples are close to adiabatic equilibrium. This is typical for the IRZ, where the residence time was long enough for the gas to (almost) completely react before being transported back to the burner mouth. The small temperature depression compared with equilibrium might be due to strain effects, temperature loss by

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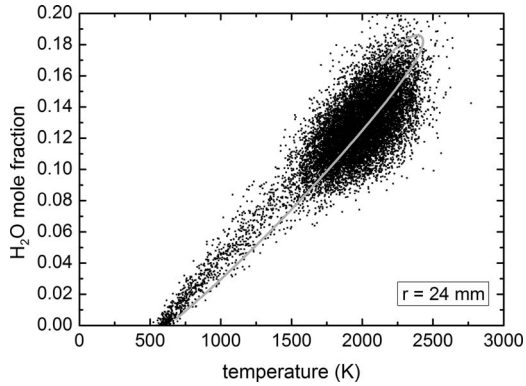

Fig. 8 Scatter plot of H_2O versus temperature in flame A at a radius of $r=24$ mm, 22 mm downstream from the burner mouth. The light gray line indicates the adiabatic equilibrium state.

radiation or contact to cold surfaces or also to measurement uncertainties. A detailed analysis of these effects would exceed the scope of this paper; however, the approach to such an analysis is, e.g., outlined in Ref. [42]. The further outwards ($r=24$ mm to $r=34$ mm) the more cold fresh gas is detected and the mean of the mixture fraction increases, as it was seen in Fig. 6. The local variation in the mixture fraction also increases, reaching from lean ($f\approx 0.02$) to rich ($f\approx 0.08$) mixtures due to a not perfect mixing of fuel and air. With increasing radius the number of samples with cold temperatures ($T\approx 630$ K) increases. Although many of them have near-stoichiometric mixture fraction values, they have not reacted yet.

In addition, a large number of samples with intermediate temperatures is seen, which reflects a state of "partially reacted." They most likely represent mixtures of hot reacted gas from the recirculation zones with fresh cold gas from the burner, which have not reacted yet due to ignition delay [42]. Other reasons for the partially reacted state might be local flame extinction or that the flame front crossed the measurement volume. However, in any case, these samples reflect effects of finite-rate chemistry. At $r=39$ mm, most samples are in cold states with a smaller mean but still strongly varying mixture fraction.

The detected hot (burned) states have a higher mixture fraction compared with the cold states. This can be explained by the higher reactivity of richer mixtures. At $r=44$ mm the mixture is even leaner but hotter in the mean. The characteristics of the scatter plots from the different radial regions are closely related to the different flow regions. Close to the flame axis, in the IRZ, the gas is mainly reacted, as explained before. Close to the ORZ ($r\geq 40$ mm), the gas is rather fuel lean and cold due to the influence of the cooling air from the burner plate. In the regions of the shear layers between the fresh gas injection and the IRZ or the ORZ, a large variation in mixing and reaction progress is present.

A representative scatter plot of the measured species concentrations is shown in Fig. 8 for H_2O in correlation with the temperature. One can see that the formation of water correlates well with the temperature, as expected, and that the results are in reasonable agreement with the equilibrium state calculation. The samples from the intermediate temperature range tend to exhibit lower temperatures as for the equilibrium. This might be due to heat loss or to effects of flame strain exerted by the high turbulence levels known from this kind of flames. The mole fractions from the other main species have also been evaluated but are not displayed and discussed here due to the limited space available.

3.2 Lifted Flame B. Flame B is a pulsating opened flame, which is stabilized farther downstream than flame A, just after the burner exit. The mean OH^* distribution is shown in Fig. 9, indicating the position of the flame brush. The images were taken with the same camera intensifier gain as for the flame A and were also averaged over 200 single shots.

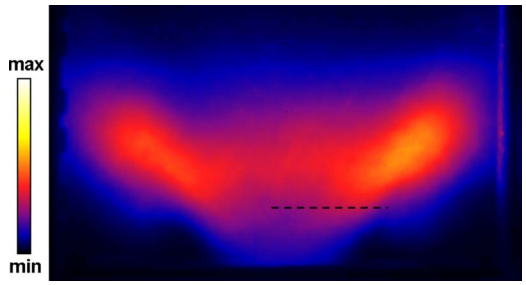

Fig. 9 Mean distribution of the OH^* chemiluminescence of flame B (averaged over 200 single shots). The dotted line indicates the position of the radial profile that was measured by 1D-Raman scattering.

In comparison to flame A, one can see that the flame zone is not so distinct and that the flame is more open and wider. The regions of the highest signal intensities lie further downstream and are less intense than in flame A.

The according OH-PLIF measurements of flame B are shown in Fig. 10, again, the mean OH distribution on top, averaged over 200 single shots, and a representative single-shot picture beneath. It is clearly visible in both images, that flame B stabilizes just outside the burner and that the ORZ contains more OH than in flame A. In the single-shot exposure it can be seen that the flame front is highly wrinkled and is also partially located in the ORZ, indicated by the spots of high signal intensities (red and white) due to the superequilibrium concentrations of the OH in the reaction zone. The contribution of the ORZ to the ignition and stabilization of the flame is therefore significantly higher than in flame A.

The radial profile of the mixture fraction f in flame B is shown in Fig. 11. As for flame A, the mean values of f are plotted against the radial position 22 mm downstream for the measuring volume in the plane of the laser sheet for the PLIF experiment. Around the burner axis (-5 mm$\leq r\leq 5$ mm), the mixture fraction is relatively high, similar to flame A. The peak value, however, is hardly dislocated from the burner axis.

In the intermediate and outer radius range (from $r=10$ mm outwards), the values of f slowly increase up to the peak value at

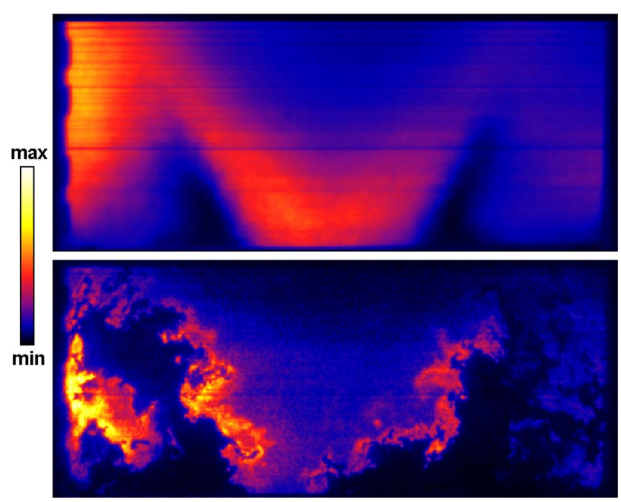

Fig. 10 OH-PLIF measurements of flame B. Top: mean OH distribution (averaged over 200 single shots). Bottom: single-shot OH-PLIF measurement.

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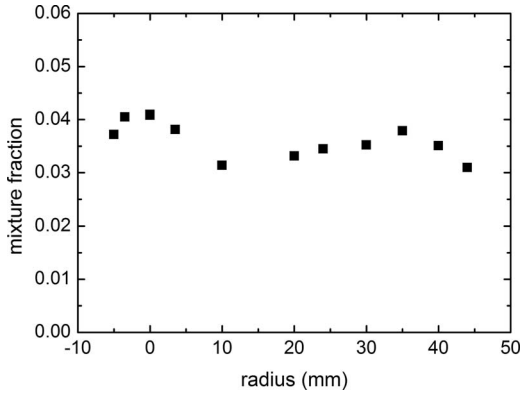

Fig. 11 Radial profile of mean mixture fraction in flame B 22 mm downstream from the burner mouth

$r=35$ mm and then decrease again in the ORZ. In comparison to flame A, the values of f are generally lower due to the higher global air equivalence ratio of flame B and also more evenly distributed, especially in the outer region ($r \geq 30$ mm).

Results of the single-shot evaluation are displayed as scatter plots in Fig. 12. The mixture fractions in all 28 measuring volumes are plotted against the temperature for different radial positions 22 mm downstream together with the equilibrium state curve (black line) and the stoichiometric mixture fraction for the NG (light gray line). As in flame A, one can see the strong variations in the mixture fraction, changing with the radial position. Partly, the measured temperature of the burned gas lies above the equilibrium curve, e.g., at $r=30$ mm. This deviation is possibly due to an inaccuracy of the measurement. However, a detailed analysis has not yet been performed and is in preparation.

In the following, the scatter plots for the different radial positions in flame B are analyzed in more detail. At $r=10$ mm, most

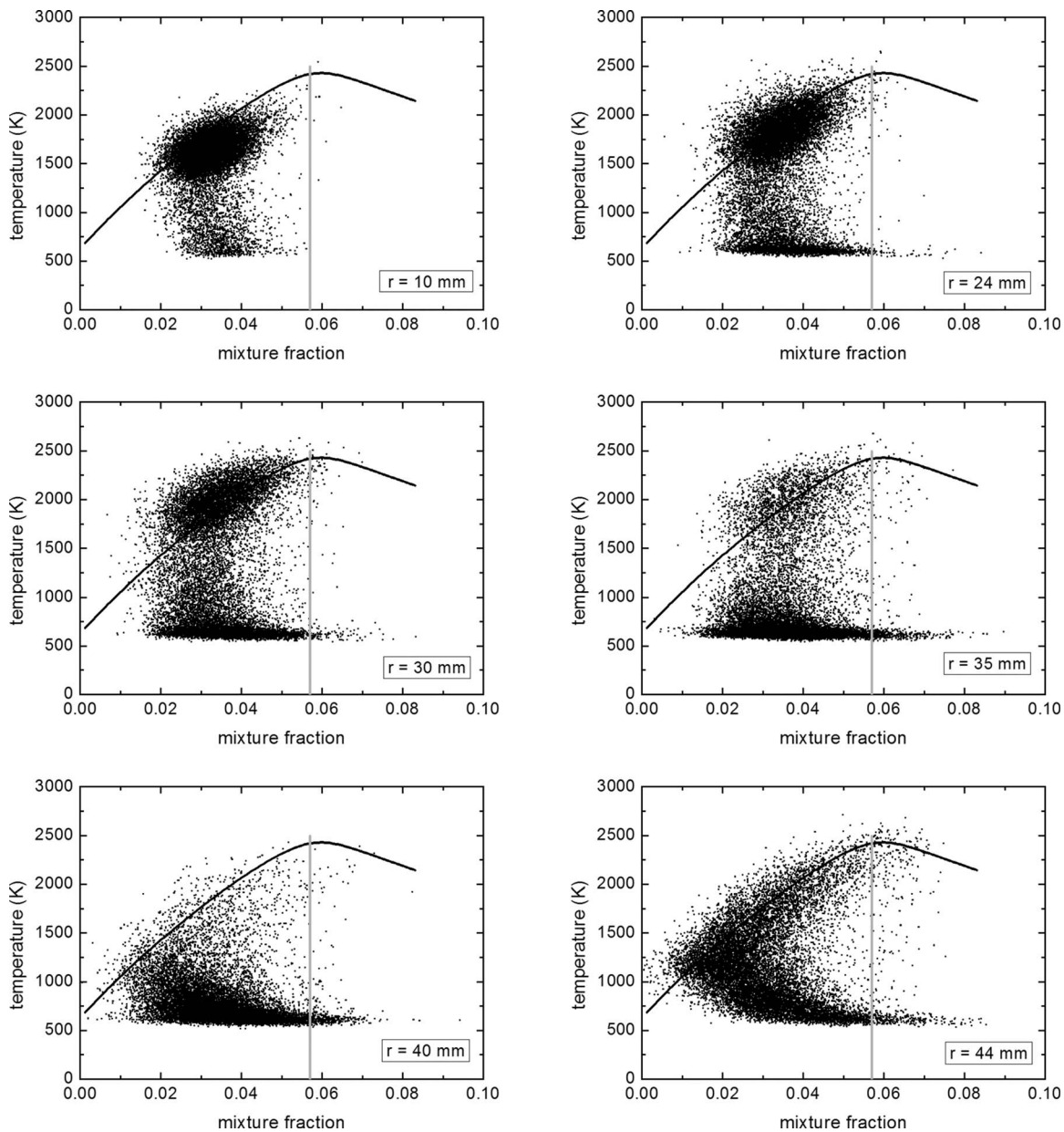

Fig. 12 Scatter plots of single-shot Raman measurements in flame B at different radial positions 22 mm downstream from the burner mouth. The adiabatic temperature (black) and the stoichiometric mixture fraction (light gray) are indicated by lines.

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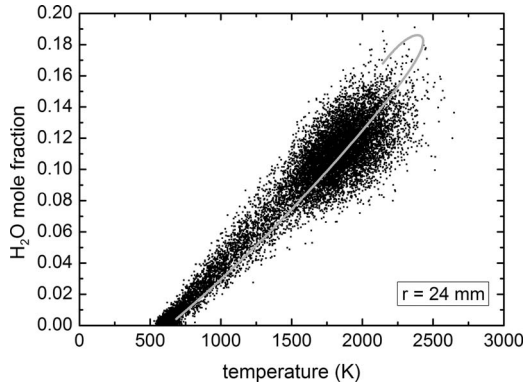

Fig. 13 Scatter plot of H_2O versus temperature in flame B at a radius of $r=24$ mm. The light gray line indicates the adiabatic equilibrium state.

measurements show high temperatures with relatively low variations in the mixture fraction. A minor part of the samples is detected at cold or intermediate temperatures, which indicates that this position is not fully inside the IRZ as it was in flame A. With increasing distance from the burner axis the variation in the mixture fraction increases due to the effects of the not perfect mixing of fuel with air, as well as the cooling air for the burner plate, which mixes with the gas in the ORZ. The portion of cold samples also increases steadily up to a radius of $r=40$ mm. At $r=44$ mm the mean temperature increases again due to the influence of the ORZ, where hot gas is transported back to the burner.

As a representative result of the single-shot evaluations of the major species, the correlation of the H_2O concentration with the temperature is shown in Fig. 13 together with the adiabatic equilibrium state. For a better comparison to flame A, the results at the same radius ($r=24$ mm) are displayed. In general, the H_2O concentration is in good agreement with the equilibrium state for the complete temperature range. The mean value of the H_2O concentration for the burned state ($T \geq 1500$ K) is smaller then in flame A, as expected, according to the lower global mixture fraction. There are also far more samples in the intermediate to cold temperature range as in flame A, where the ignition has not set in yet, or the gas is only "partially" burned due to the afore mentioned reasons.

4 Summary

A downscaled industrial gas turbine burner was equipped with an optical combustion chamber and installed in a high-pressure test rig. It was operated at different flame conditions at elevated pressure. Laser spectroscopic and optical measurements have been performed to investigate the flame behavior and to provide data for validation purposes of numerical simulations. Planar OH-LIF and OH^* chemiluminescence were applied to reveal the structure of the reaction zones and hot regions and the position and shape of the flame brush, respectively. 1D-laser Raman scattering has been applied to measure the major species concentrations, the temperature and the mixture fraction along a line of 7 mm length. The application of this technique to a high-pressure test rig of this size was challenging and problems related to the high laser intensities, and stained windows had to be overcome. Two different flames in terms of shape and stabilization (flames A and B) were selected and investigated in more detail. The OH-PLIF and OH^* chemiluminescence measurements revealed that flame A, which was stabilized within the burner cone, has a smaller and more distinct flame zone than the lifted and more unstable flame B. The single-shot Raman measurements enabled a detailed characterization of the mixing and reaction progress of these flames. Although it was known before that the degree of premixing in GT burners is limited, it could now be quantified for the first time without spatial or temporal averaging. In both flames, the mixture fraction exhibited a significant scatter at a distance of $\sim 1/3$ of the burner exit diameter, and its distribution varied with the radial position. The scatter plots also revealed that the thermochemical state of the flames varied from nonreacted via partially reacted to fully burned. The partially reacted mixtures had intermediate temperatures and stemmed probably from the mixing of hot burned gas with cold fresh gas before ignition has set in. The characteristics of the scatter plots from different radial regions were in accordance with the different flow field regions, i.e., the inflow of fresh gases, the recirculation zones, and the shear layers.

Acknowledgment

The authors would like to thank A. Koch, R. Sadanandan, and their colleagues from the DLR test rig for their support and contributions to the investigations. The financial support from the State Baden-Württemberg, the EnBW Energie Baden-Württemberg AG and ALSTOM Power within the research program KW21 is gratefully acknowledged.

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